**MADHAVI PAGARE**

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## RESEARCH INTERESTS

### Causal Analysis in Social Determinants of Mental Health for Opioid Use Disorder

My research focuses on unveiling causal relationships between Social Determinants of Mental Health (SDMHs) and Opioid Use Disorder (OUD) outcomes using advanced frameworks. By integrating the Multitask Multilabel Clinical-Longformer (MMCL) for SDMH detection and Siamese Neural Networks (SNN) for subgroup discovery, I aim to address the limitations of traditional correlation studies. This approach enables precise causal effect estimation, guiding targeted interventions and improving OUD care through insights derived from unstructured clinical notes

### Predicting Mental Health Determinants Through Lived Experiential Essays

I explore the intersection of education and mental health by employing advanced AI techniques to predict Social Determinants of Mental Health (SDMHs) from students' experiential essays. Utilizing a Human-LLM Interaction for Annotation (HLIA) framework and a Multitask Cascaded Neural Network (MTCNN), my work highlights the potential of Large Language Models in annotating and predicting SDMHs. This research bridges educational data mining and mental health, addressing data limitations while creating scalable, automated solutions for understanding student experiences

### Enhancing Opioid Care Through NLP and Human-LLM Collaboration

My research investigates the integration of Natural Language Processing (NLP) and human oversight to extract Social Determinants of Mental Health (SDMHs) from clinical notes of Opioid Use Disorder (OUD) patients. By leveraging the Human-in-the-Loop-Language Model Interaction for Annotation (HLLIA) framework and Multilevel Hierarchical Clinical-Longformer Embeddings (MHCLE), I aim to improve annotation reliability and model performance. This work emphasizes the necessity of combining human expertise with AI to refine healthcare outcomes and inform patient care strategies.

## Academic & Professional Experience

**Texas A&M University-Corpus Christi** Nov 2025-Current

Postdoctoral Research Associate

**Merrimack College**

Research Collaborator Spring 2024-June 2025

**University of Massachusetts Lowell**

Teaching Assistant — Fall 2022-Summer 2025

Research Assistant — Summer 2023

**LivedX**

Machine Learning Intern — Summer 2023

Research Collaborator — Fall 2023

**UMass Lowell, Civil and Environmental Engineering**

Research Assistant — March 2023

**University of Colorado, Anschutz medical campus**

Student Research Assistant: EMR-based research — Spring 2022-Summer 2022

**University of Colorado, Denver**

Graduate Assistant: I.T Lab — Fall 2021-Spring 2022

Teaching Assistant — Spring 2021

**Veermata Jijabai Technological Institute, Mumbai**

Assistant Professor — July 2017-Dec 2018

## EDUCATION

**University of Massachusetts Lowell**
Ph.D. in Computer Science (August'2025)
GPA: 3.5/4.0

**University of Colorado, Denver**
Master of Science (Computer Science)
GPA: 3.71 /4.0 (Jan'2021-June'2022)

**University of Mumbai**
Master of Engineering (Computer Engineering)
First Class with Distinction

## Professional Experience

**Postdoctoral Research Associate** — **Texas A&M University-Corpus Christi** — **Nov 2025-Present**

- Conducting research on advanced AI and machine-learning models with a focus on time-series forecasting, modeling, and anomaly detection using deep-learning architectures.

- Developing knowledge-distillation and model-compression techniques to enhance the efficiency and scalability of neural networks for scientific data analysis.

- Collaborating with faculty and students to apply data-driven statistical methods in interdisciplinary research, contributing to manuscripts and future grant proposals.

**Research Collaborator** **Merrimack College** **April 2024-June 2025**

**<u>Opiod Use Disorder Project</u>**

- Engineered a Multitask Multilabel Clinical-Longformer (MMCL) model, achieving 99.00% accuracy, 98.99% F1-score, and 99.02% precision in classifying Social Determinants of Mental Health (SDMHs) from clinical notes in the MIMIC-IV dataset, enhancing NLP-driven medical analytics.
- Pioneered a Siamese Neural Network (SNN)-based subgroup discovery framework, improving causal effect estimation of SDMHs on Opioid Use Disorder (OUD), refining treatment impact analysis in large-scale clinical datasets.
- Developed and curated the SDMH-OUD-Clinic dataset, preprocessing and structuring 2,636 clinical discharge summaries, facilitating high-precision causal inference analysis for healthcare policy recommendations.
- Led an extensive causal inference study leveraging propensity score matching and bootstrap sampling, demonstrating:
    - Healthcare Handover (HCH) reduced OUD risk by 48.4% (ATE = -0.483886, $p < 0.05$)
    - Obstacles to Medical Care (OMC) increased OUD risk by 41.4% (ATE = 0.413967, $p < 0.05$)
- Optimized SDMH text representation using Bio_ClinicalBERT embeddings, leading to more precise detection of OUD risk factors and enhancing clinical decision-making in AI-driven healthcare applications.
- Validated the robustness of the MMCL model on Infant Health and Development Program (IHDP) data, outperforming baseline causal inference models on Precision in Estimating Heterogeneous Effects (PEHE), ensuring real-world applicability in personalized healthcare interventions. (**IEEE-BHI Conference Accepted**)

**<u>Breast Cancer Recurrence Project</u>**

- Tackled the need to understand how Sociocultural Factors impact breast cancer recurrence in an area often limited to correlation-based insights.
- Designed a robust framework to estimate the causal influence of 22 Sociocultural Factors using real-world patient data. Developed a transformer-based Clinical Longformer Multi-Task Multi-Label Classifier to extract Sociocultural Factors from unstructured clinical notes.
- Integrated three causal inference methods-Siamese Neural Network-based subgroup discovery, Causal Effect Variational AutoEncoder, and Inverse Probability Treatment Weighting with Regression Adjustment within a unified pipeline to compute Conditional Average Treatment Effects (CATE).
- Revealed nuanced causal relationships between Sociocultural Factors and cancer recurrence, delivering actionable insights for personalized treatment and healthcare policy optimization. (**JMIR under Review**)(**Abstract Accepted -IEEE-BHI 2025**)

**NLP Researcher** **University of Massachusetts Lowell** **Aug 2023-June 2025**

- Engineered a hierarchical text classifier for clinical notes in Collaboration, adeptly identifying opioid abuse with a substantial first-tier accuracy of 99.00%, F1-Score: 98.99%, precision: 99.02%, Recall: 98.97%.
- Executed a binary classification in the second tier, effectively discerning the presence of social determinants of mental health in clinical reports with a notable accuracy of 95.82%, F1-Score: 95.68%, Precision: 95.85%, Recall:95.80%
- Advanced to a sophisticated multi-label classification approach in the third tier, accurately identifying distinct social determinants of mental health with an accuracy of 96.29%, F1-Score:95.41%, Precsion:96.03, Recall:94.81%
- Optimized the performance of a Roberta model on clinical data, delivering comprehensive classification outcomes in a condensed GPU computation window of 5 hours and 15 minutes. (**JAMIA open Accepted**)

**Machine Learning Intern** **LivedX** **May 2023 - Jul 2023**

- Worked on developing the Human-LLM Interaction for Annotation (HLIA) method to label texts (text analytics) with information on mental health determinants in Collaboration. Prepared 13 Social determinants of Mental Health. Leveraged ChatGPT.
- Developed in team a Multitask Cascaded Neural Network (MTCNN) classification algorithm to anticipate these variables in students' data.
- Investigated the use of large language models (LLMs) in the creation of a mental health determinant prediction system using student's data.
- Leveraged MTCNN classification method and the HLIA framework to improve annotation and prediction processes. After training and testing on a 70-30 data split, reported the performance metrics: Accuracy: 92%, F1-Score: 92%, Precision: 92%, Recall: 97%. (**EDM Conference Accepted**)

**Research Assistant** **University of Massachusetts Lowell** **Jul 2023 - Sep 2023**

- Developed a sequential sentence classification model for analyzing social media posts in the context of dementia care, applying text analytics to extract meaningful insights.
- Applied data augmentation and fine-tuned pre-trained language models, enhancing classification accuracy and model generalization.
- Trained and tested the model on datasets containing posts from family members, friends, and caregivers, achieving 92.43% classification accuracy.
- Automated dementia care post classification, enabling scalable AI-driven predictive modeling to support caregiving insights and intervention strategies.

**Research Assistant** **University of Massachusetts Lowell** **Jan 2023-March 2023**
**(Data Science Researcher-NSF-Funded project)**

- Developed a Python-based automation tool to extract and process 308 property attributes from Zillow using APIs and web scraping, leveraging pandas and requests to support a Graph Neural Network (GNN)-based water contamination analysis system.
- Designed data structuring workflows to ensure seamless integration into a graph-compatible analytical framework.
- Collaborated with cross-disciplinary researchers to validate data quality and alignment with project objectives, enhancing scalability and precision.
- Delivered processed datasets for an NSF project update, showcasing advancements in AI-driven data preprocessing techniques. (**Frontiers in Environmental Engineering Journal Accepted**)

**Software Engineer Intern** **Eitacies Inc** **Jun 2022 - Jul 2022**

- Analyzed datasets using statistical modeling techniques (e.g., standard deviation, regression analysis) to uncover trends for predictive analytics.
- Built and optimized MongoDB-based ML pipelines to process and store facial emotion recognition data, improving model performance monitoring.

**Student Assistant (EMR-Research)** **University of Colorado, Anschutz, US** **Apr 2022 - Jul 2022**

- Extracted and organized diabetes and device data from Epic EMR, driving accuracy and efficiency in workflows.
- Maintained high-quality datasets through precise REDCap entry and proactive data validation.
- Leveraged Excel/CSV tools to streamline reporting and improve accessibility for cross-functional teams.

**SQL Database Administrator** **Goalmine Corporate Services, LLP, India** **Jan 2019 - Dec 2020**

• Administered SQL databases, optimizing performance and storage efficiency, reducing query execution time by 70%
• Implemented backup and recovery solutions for SQL Server 2008, 2008 R2, and 2012, ensuring 99.9% data availability.
• Optimized large-scale datasets in Healthcare, Banking, and Financial domains, improving query performance and data retrieval speeds.
• Configured SQL jobs, user access controls, and audits, strengthening database security and regulatory compliance.
• Enhanced database performance in AWS and Azure cloud environments, reducing latency and improving processing efficiency.

**Teaching Experience**

**Teaching Assistant** **University of Massachusetts Lowell** **Sep 2022-May 2025**

- Assisted in labs for Computing I, Computing III, and Computing IV, supporting students in C++ programming from introductory to advanced levels.
- Guided students through hands-on programming exercises on arrays, data structures, object-oriented design, inheritance, polymorphism, templates, and software engineering principles.
- Provided technical support with Unix shell, GCC (g++), Makefiles, and Ubuntu Virtual Machine for C++ development and debugging.
- Mentored students on 500+ line projects, including the use of SFML (graphics) and Boost libraries.
- Helped evaluate assignments, quizzes, and portfolios, ensuring students met course learning outcomes.

**Teaching Assistant** **University of Colorado Denver** **Spring'21**

- Installed and configured XAMPP software, establishing a strong foundation for hands-on database management exercises.
- Conducted virtual office hours via Zoom, assisting students in Normalization, Relational Models, ER Models, Relational Mapping, Relational Algebra, EER Models, File Organization, Indexing, Database Security, Data Mining, and Distributed Databases.
- Evaluated and graded assignments and quizzes, providing timely feedback to support student progress.
- Prepared detailed quiz solutions, reinforcing key database concepts and ensuring clarity in assessments.

**Assistant Professor** **Veermata Jijabai Technological Institute (VJTI), Mumbai** **July 2017 - Dec 2018**

- Delivered courses in Software Architecture, Service-Oriented Architecture, Software project management at undergraduate and graduate levels.
- Designed course materials, conducted lectures, and supervised lab sessions.
- Advised students on research projects and academic development.

**Lecturer and Assistant Professor** **YTCEM Mumbai** **July 2009 - June 2017**

- Designed and delivered undergraduate courses in C, C++, Java, Database Management Systems, Computer Networks, AI, and Image Processing, enhancing student learning outcomes.
- Mentored and guided 10+ undergraduate projects and assisted in 3+ M.Tech research projects, fostering innovation and critical thinking.

- Spearheaded placement activities as Placement Coordinator (2011–2014), establishing industry partnerships and improving student employability.
- Directed and managed the NBA accreditation process, compiling placement reports and implementing academic improvements to enhance program quality.

**Technical Skills**

- **Programming Language Skills:** C, C++, Java, Python, data structures, algorithm design, Digital Signal Processing, Audio Processing
- **Database/Backend:** MS SQL server 2008, 2008 R2, 2012, T-SQL, HTML, SQL, SQL Server Installation, Database Management
- **Business Intelligence/Data Analytics tools/libraries/platforms:** Microsoft Excel, NumPy, pandas, Scikit-Learn, Matplotlib, Seaborn, Microsoft Access, PyTorch, TensorFlow, Git, Artificial Intelligence, NLP, Generative AI, NLTK, spaCy, Text Recognition, statistics, reinforcement learning, data cleaning, data preparation, data manipulation, solving real-world business problems, generative models, modeling and visualization, statistical data analysis, NumPy, SciPy, SciKit, Spark, Data Modeling, Data Visualization, Data Mining, Sklenar Clinical-BERT, Clinical Longformer ,Clinical Bigbird, Bio_ClinicalBERT, BERT, RoBERTa, linear algebra
- **Cloud platform:** Amazon Web Services (AWS), Microsoft Azure, Google Colab. Proficient in CUDA programming for GPU acceleration
- **Certification:** Introduction to Oracle 9i: SQL (90.5%), Data Science Math Skills-Coursera, Hugging Face
- **Related Coursework: (Graduate/PhD Level)**
  Algorithms • Data Communications I • Internet of Things • Advanced Computer Architecture • Operating Systems • Cyber-Physical Systems • Theory of Automata • Machine Learning • Deep Learning • Advanced Social Computing • Applied Graph Theory • Social Networks and Informatics • Software Project Management • Computer Science Research • Parallel Computer Architecture • Algorithms and Complexity • Object-Oriented Analysis and Design • Network Protocol and Networking • Neural Networks and Fuzzy Systems • Software Engineering • Distributed Operating System • Advanced Database Management Systems • Image Processing • E-Commerce

*(University of Colorado Denver - M.S. in Computer Science, University of Massachusetts Lowell - Ph.D. in Computer Science, and University of Mumbai - M.E. in Computer Engineering)*

**Research Publications & Conferences**

**Conference Proceedings**

1. **Madhavi Pagare**, Inyene Essien Aleksi, Atqiya Munawara Mahi, **Mohammad Arif Ul Alam**, Finding Causal Impacts of Social Determinants of Mental Health on Opioid Use Disorder from Clinical Notes, The IEEE-EMBS International Conference on Biomedical and Health Informatics, 2024. (**IEEE-BHI Accepted**)
2. Alam, Mohammad Arif Ul, **Madhavi Pagare,** Susan Davis, Geeta Verma, Ashis Biswas, and Justin Barbern. "Empowering Predictions of the Social Determinants of Mental Health through Large Language Model Augmentation in Students' Lived Experiential Essays." In Proceedings of the 17th International Conference on Educational Data Mining, pp. 617-627. 2024. (**EDM Accepted**)

**Peer-Reviewed Journals**

1. **Madhavi Pagare,** Inyene Essien-Aleksi, Deva Sai Kumar Bheesetti, and Mohammad Arif Ul Alam, "Augmenting Large Language Models to Predict Social Determinants of Mental Health in Opioid use Disorder Using Patient Clinical Notes" (**JAMIA open Accepted**)

2. **Madhavi Pagare**, Inyene Essien Aleksi, Mohammad Arif Ul Alam, "Cancer Does Not Strike Alone: How Social Determinants of Mental Health Shape Breast Cancer Recurrence, The IEEE-EMBS International Conference on Biomedical and Health Informatics (**JMIR** under Review)(**Abstract Accepted at IEEE-BHI 2025**)
3. Anaadumba, Raphael, Yigit Bozkurt, Connor Sullivan, **Madhavi Pagare**, Pradeep Kurup, Benyuan Liu, and Mohammad Arif Ul Alam. "Graph Neural Network-based Water Contamination Detection from Community Housing Information." Frontiers in Environmental Engineering 4: 1488965. (**Frontiers in Environmental Engineering Accepted**)
4. Irankhah Elyas,**Madhavi Pagare,**Wolkowicz, Kelilah,Chetla, Lokesh, Mohammad Arif Ul Alam **"**Machine Learning–Enhanced Causal Inference of Surgical Decisions and Rehabilitation Strategies in Traumatic Brain Injury" (**Frontiers in Neurology Accepted**)
5. Yigit Bozkurt, Raphael Anaadumba, **Madhavi Pagare,** Mohammad Arif Ul Alam, Pradeep Kurup "Critical Review for Lead Detection in Water Distribution Systems Using Machine Learning".(submitting soon)
6. Deva Sai Kumar Bheesetti, Mohammad Arif Ul Alam, **Madhavi Pagare "**Opioid use Disorder ordinal Classification" (target JAMIA)
7. Savita Chavan , Mangal Sonawane, Ashwini Chavan, **Madhavi Sarjare(Madhavi Pagare)** (2012). Middleware: An Architecture for Distributed Systems Services. in International journal of Engineering Science and Innovative Technology( IJESIT), ISSN: 2319-5967, VOLUME 1,ISSUE 2, NOVEMBER 2012.
8. **Madhavi. K. Sarjare** (**Madhavi Pagare**), S.L.Vaikole "Web-scale information Extraction from unstructured and ungrammatical data sources", International Journal of Computer Science Engineering and Information Technology Research (IJCSEITR), ISSN (P): 2249-6831; ISSN (E): 2249-7943, Vol. 4, Issue 2, Apr 2014, 109-114.
9. **Madhavi. K. Sarjare(Madhavi Pagare)**, S.L.Vaikole "Scaling the Information Extraction from Unstructured and Ungrammatical Data Sources on Web", International Journal of Engineering and Science (IJES) ISSN: 2319 – 1813, ISBN: 2319 – 1805, Volume 3 ,Issue 6 Pages 36-45 ,2014.

**Mentorship**

**Current Advisees**

## University of Massachusetts Lowell:

- **Deva Sai Kumar**, M.S. Student (2024 - present)
- **Venkataraman Potluri,** MS Student (2024-present)
- **OConnor, Thomas D,** MS Student (2024)
- **Chetla Lokesh,** MS Student (2025-present)
- **Raphael Anaadumba**, Ph.D. CS (2023 - present)
- **Yidong Zhu**, Ph.D. CS (2023 - present)
- **Irankhah Elyas**, Ph.D. Mech (2025-present)

## University of Colorado Denver:

- **Susan Davis**, Ph.D. EdD (2023)

**Past Advisees**

**VJTI, Mumbai:**

- **Vaibhav Ingale, Vishaka Mohod, Mayur Jadhav** [M.Tech Students-2018)

**YTCEM, Mumbai:**

- **Rinita Ghagare, Priyanka Solunkhe, Shrikant Walunj** [Undergraduate Student-Computer Engineering (2013)]

## ACADEMIC PROJECTS

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**Micro credential Prediction for student's experiential essays** (Deep Learning, NLP) July 2023-Oct 2023

- Developed and fine-tuned a RoBERTa-based NLP system, achieving a 93.06% accuracy in micro-credential evaluation.
- Revolutionized skill recognition with a multi-tiered classification strategy, elevating the assessment of diverse educational backgrounds.

**Personalized Cancer diagnosis (Machine Learning, Python)** July 2021-Dec 2021

- Designed an early Cancer prediction System that ensured initial Breast cancer.
- Implemented the Naive Bayes, K-Nearest Neighbor, logistic Regression, linear support vector machines, and Random Forest classifier for breast Cancer project based on machine learning prediction algorithm resulting in an accuracy of 94%.

**Hospital Emergency Management System (Java, Python, HTML, CSS, JavaScript, JQuery, Bootstrap, Firebase db)** July 2019-Dec 2019

- Developed and implemented a user-friendly android application for managing Patient records. Here the application is classified into the patient, Doctors, and admin depending on different privileges. RFID cards are used for tracking and collecting patient's information.

## LEADERSHIP SKILLS AND AFFILIATIONS

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|---|---|---|
| **Reviewer** | **IEEE-EMBS International Conference 2025** | **July 2025** |
| **Invited Presenter (Poster and Lecture)** | **IEEE-EMBS International Conference** | **Nov 2024** |
| **Poster Presenter** | **Manning School of Business \| UMass Lowell** | **Sep 2024** |
| **Mentorship (Graduate Students)** | **University of Massachusetts Lowell** | **March 2024** |
| **Teaching and Research Assistant** | **University of Massachusetts Lowell** | **Sep 2022 - May 2025** |
| **Member and Reviewer** | **7th IEEE International Symposium on Smart Electronics System** | **Sept 2021 - Present** |
| **Graduate Assistant (I.T)** | **University of Colorado, Denver** | **Aug 2021 - May 2022** |

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|---|---|---|
| **Teaching Assistant** | **University of Colorado, Denver** | **Jan 2021 - May 2021** |
| **Member** | **Indian Society for Technical Education (ISTE)** | **2012** |
| **NBA Coordinator** | **Veermata Jijabai Technological Institute (VJTI),India** | **2017-2018** |
| **Placement Coordinator** | **YTCEM,Mumbai** | **2011-2014** |
| **Organized Guest Lecture** | **Veermata Jijabai Technological Institute (VJTI),India** | **2017-2018** |

**International Certifications**

- **LinkedIn**
  - Applied AI: Building NLP Apps with Hugging Face Transformers - Jan 2024
  - Applied AI: Getting Started with Hugging Face Transformers - Jan 2024
- **Microsoft**
  - Microsoft Certified Azure Data Fundamentals (DP-900) – Feb 2022
    - Credential ID: l134-6027
  - Microsoft Certified Azure Fundamentals (AZ-900) – Jan 2022
    - Credential ID: l126-5741
- **Duke University**
  - Data Science Math Skills – Jul 2020
    - Credential ID: JSNWFYGVGVPF
- **Oracle**
  - Introduction to Oracle9i: SQL – Sep 2008
    - Credential ID: QBBSYD5721

**Workshops & Faculty Development Programs Attended**

- Big Data & Hadoop Training - MCAL Global (July 2014)
- 1-Week ISTE Workshop on Database Management Systems - IIT Bombay (December 2010)
- AICTE-QIP Sponsored Workshop on Spatial Data & Data Mining - VJTI, Mumbai (March 2018)
- Global Initiative for Academic Networks (GIAN) Workshop on Trustworthy AI - VJTI, Mumbai (December 2016)

- 2-Day Weekend Training Program on Machine Learning - IIT Kharagpur (December 2017)
- International Faculty Development Program (QIP-SPPU, Pune) on Data Science Applications - Pimpri Chinchwad College of Engineering, Pune (February 2018)
- 1-Week Workshop on ICT-Based Teaching Methodologies - YTIET, Mumbai (July 2013)
- IBM Certified Training on Object-Oriented Analysis & Design using UML (January 2011)
- Oracle University Training: Developing and Distributed Applications using VB 6.0 - NIIT (December 2005)

Sincerely,

Madhavi Pagare